

August 8. Eisenstein Criteria.



We have proven $\mathbb{Z}[X]$ is a UFD.

and have shown that the irreducibles in $\mathbb{Z}[X]$ are the primes in $\mathbb{Z}$ and the primitive ones that are irreducible in $\mathbb{Q}[X]$. Today we give some useful irreducibility criteria.

eg. Consider $\sqrt[3]{2}$ is a root of

$x^3 - 2$. (why? set $x = \sqrt[3]{2}$.

then $x^3 = 2$ which is $x^3 - 2 = 0$).

Is $x^3 - 2$ irreducible?

Well. The first question should be where is it irreducible?

In $\mathbb{R}[X]$. it is reducible for it factors

as $(X - \sqrt[3]{2})$, which is an irreducible

factor as it is a deg 1 polynomial in $\mathbb{R}[X]$.

a polynomial ring over a field. , divides $X^3 - 2$

for $\sqrt[3]{2}$ is a root of $X^3 - 2$ in $\mathbb{R}$.

As $\mathbb{R}[X] \subseteq \mathbb{C}[X]$, any factorization in $\mathbb{R}[X]$

also appears in $\mathbb{C}[X]$. So $X^3 - 2$ is reducible

in $\mathbb{C}[X]$. Note $X^3 - 2$ is primitive in $\mathbb{Z}[X]$.

So it is irreducible in $\mathbb{Z}[X]$ iff

it is irreducible in $\mathbb{Q}[X]$. Since $\mathbb{Q}$ is a field and $\mathbb{Q}[X]$ is ED, it is easier to check irreducibility in $\mathbb{Q}[X]$.

Note $x^3 - 2$ is of deg 3. So it is irreducible in $\mathbb{Q}[X]$ iff it has no roots in $\mathbb{Q}$. Now it is easy to see the roots of $x^3 - 2$ are $\sqrt[3]{2}, \sqrt[3]{2}\omega, \sqrt[3]{2}\omega^2$ where ω is a 3rd-root of unity in particular $\omega = e^{i2\pi/3}$

Then clearly no root lies in $\mathbb{Q}$.

If you don't know about roots of unity.

recall the rational root test, which says

any rational root must be within $\pm 2, \pm 1$

where clearly none is a root.

So $x^3 - 2$ is irreducible in both

$\mathbb{Z}[x]$ and $\mathbb{Q}[x]$. (after a bit of work).

But now $\sqrt[5]{2}$ is a root of $x^5 - 2$

Is it irreducible?

We need new tools!

Theorem 1 : (Eisenstein Criterion).

Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

for $n > 1$ in $\mathbb{Z}[x]$. Suppose there

exists a prime number p such that.

$$(1) \quad p \nmid a_n.$$

$$(2) \quad p \mid a_i \quad \forall \quad 0 \leq i < n$$

$$(3) \quad p^2 \nmid a_0.$$

Then it is impossible to write $f(x)$ as

$g(x)h(x)$ for $g(x), h(x) \in \mathbb{Z}[x]$ and

$\deg g(x), \deg h(x) < \deg f(x)$.

proof: Suppose $f(x) = g(x)h(x)$ for

some $g(x), h(x) \in \mathbb{Z}[x]$ such that

$$\deg g(x), \deg h(x) < \deg f(x) = n.$$

$$\begin{aligned} \text{Write } g(x) &= \sum_{u=0}^s b_u x^u && \text{for } s < n. \\ h(x) &= \sum_{v=0}^t c_v x^v && \text{and } t < n \end{aligned}$$

The constant term of $f(x) = g(x)h(x)$

is $b_0 c_0$. Since $p \mid b_0 c_0$ and p is

a prime, then $p \mid b_0$ or $p \mid c_0$.

We claim p does not divide both b_0 and c_0 .

Indeed, otherwise $p^2 \mid b_0 c_0 = c_0$

Without the loss of generality.

assume $p \mid b_0$ and $p \nmid c_0$

We prove by induction that $p \mid b_u \forall 0 \leq u \leq s$

The base case is proven.

Assume $p \mid b_u \forall 0 \leq u \leq k-1$ for $k \leq s < t$

We show $p \mid b_k$.

Consider $a_k = \sum_{u+v=k} b_u c_v$

$$= b_0 c_k + b_1 c_{k-1} + \dots + b_{k-1} c_1 + b_k c_0$$

$$= \sum_{\substack{u+v=k \\ u < k}} b_u c_v + b_k c_0$$

Note that $p \mid a_k$ and $p \mid \sum_{\substack{u+v=k \\ u < k}} b_u c_v$

So $p \mid b_k c_0$ which implies

$p \mid b_k$ or $p \mid c_0$. But $p \nmid c_0$

So $p \mid b_k$.

By induction, we conclude $p \mid b_u$

$\forall 0 \leq u \leq S$. In particular,

we can factor $g(x) = p \cdot g'(x)$.

Then $f(x) = g(x)h(x) = p \cdot g'(x)h(x)$.

and p divides all coefficients of $f(x)$.

which is a contradiction.

